# ITNet: Low-Shot Instance Transformation Network for Weakly Supervised Object Detection in Remote Sensing Images

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***Abstract*— Several studies of weakly supervised learning have been applied to object detection in remote sensing images (RSIs), while critical challenges like part domination and class confusion remain, which lead to poor accuracy compared with fully supervised object detection tasks (e.g., FRCNN and YOLOv4) and natural images set tasks (e.g., PASCAL VOC 2007). The model is prone to focus on the most discriminate part of the object due to the fact that image-level annotations are lack of instances' box information. Moreover, class confusion arises when the model is attempted to recognize instances of different categories that consistently coexist within a single training sample. To address the problems, we developed the low-shot instance transformation network (ITNet). ITNet is able to transform part domination boxes and misidentified confusion classes to be more accurate, which is trained with a combination of a small number of strong annotations and weak annotations. First of all, elastic cluster selection (ECS) is proposed to mine high-quality weak pseudoannotations from the output of only weakly supervised object detection (WSOD) models [e.g., online instance classifier refinement (OICR)]. Label reassignment (LRA) allows the correction of weak pseudoannotations with category noise by the recognition knowledge learned in strong annotations to alleviate two-class confusion. Then, semisupervised elastic match (SSEM) is employed to update the annotations to make full use of strong annotations. Comprehensive experiments are carried out on NWPU VHR-10.v2 and DIOR, proving that the proposed ITNet outperforms the previous state-of-the-art significantly.**

***Index Terms*— Low-shot weakly supervised object detection (LS-WSOD), remote sensing images (RSIs), weakly supervised object detection (WSOD).**

## NOMENCLATURE

| | |
|---|---|
| $\mathcal{I}_\alpha$ | Training set for ITNet in mixed path $\alpha$. |
| $\mathcal{I}_\beta$ | Training set for ITNet in weak path $\beta$. |
| $b_w$ | Weak pseudoannotations obtained from OICR. |
| $b'_w$ | Transformed weak pseudoannotations obtained from correction branch. |
| $b^k_{sp}$ | Semipseudoannotations at training stage $k$ obtained by SSEM from stage $k-1$. |
| $b_s$ | Manually labeled strong annotations. |
| $L$ | Total loss of ITNet. |
| $L_{\alpha C}$ | Correction branch loss in mixed path $\alpha$. |
| $L_{\alpha R}$ | Refinement branch loss in mixed path $\alpha$. |
| $L_{\beta C}$ | Correction branch loss in weak path $\beta$. |
| $L_{\beta R}$ | Refinement branch loss in weak path $\beta$. |
| $R_e$ | Score elastic rate, parameter for ECS. |
| $I_e$ | Cluster IoU threshold, parameter for ECS. |

Manuscript received 28 May 2023; revised 24 August 2023 and 14 November 2023; accepted 16 December 2023. Date of publication 16 January 2024; date of current version 31 January 2024. This work was supported by the Youth Innovation Promotion Association, Chinese Academy of Sciences (CAS), Beijing, China. *(Corresponding author: Zongxu Pan.)*

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Digital Object Identifier 10.1109/TGRS.2024.3354999

## I. INTRODUCTION

IN THE past decade, weakly supervised object detection (WSOD) has been widely adopted in natural vision and remote sensing image (RSI) tasks, which can avoid extensive manual labeling and take full advantage of image-level annotations. Research on natural vision has focused on the application of multiple instance learning (MIL) methods to deep networks to address issues such as component dominance and augmentation consistency [1], [2], [3], [4], [5]. The growing success of the research in the traditional vision field and the data characteristics of the RSIs have also given rise to the rapid development of the research in the field of RSIs. However, there is still a huge accuracy gap between weakly supervised methods and fully supervised methods, so the idea of low-shot weakly supervised learning was brought up [6], [7]. The low-shot WSOD (LS-WSOD) models aim to improve the accuracy by using a combination of strong and weak labels. Such methods, which can boost detection accuracy by adding a small amount of strongly supervised data, are a new emerging interest in recent research.

Among them, box correction network (BCNet) designed a two-path, two-branch structure, trying to exploit the information from the strong and weak annotations, thus obtaining the ability to correct from the part-dominated weak pseudoannotations to the complete strong annotations [6]. However, BCNet did not properly address the following three issues. First, the weak pseudoannotations are selected from the output of online instance classifier refinement (OICR) [4] by the hard threshold method, while the candidate regions and outputs of the weakly supervised network are clustered. Namely, there are many high-scoring prediction boxes clustered together in the regions that are likely targets, and the highest scored prediction box is often not the one that covers the target most. Second, if there are two classes of objects always present

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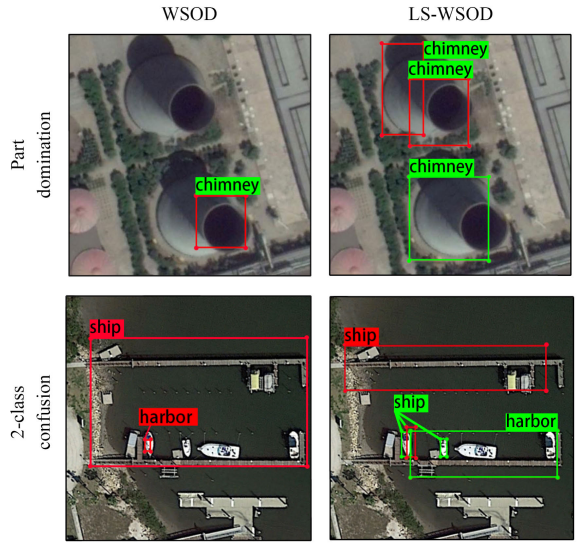



Fig. 1. Part domination and class confusion are two major problems that WSOD and LS-WSOD methods would encounter, which causes significant reduction in model performance. Part domination appears for WSOD is lack of boxes' annotation, thus making model focus on the most discriminative part of objects. Two-class confusion in WSOD appears when two kinds of categories exist in one weak annotation at the same time, which can prevent the model from identifying the class of the objects.

in the same positive bag at the same time, then the weakly supervised network cannot learn the recognition information of both classes correctly. Such a situation is especially common in RSIs. For example, if ships and harbors tend to appear in an RSI at the same time, the trained weakly supervised network cannot correctly predict the categories of the ship and harbor instances, which is called class confusion in WSOD. It should be acknowledged that class confusion can also arise in situations where multiple classes possess highly similar features, which is particularly common in the field of domain adaptation. In this work, class confusion is discussed in the context of WSOD, where instance-level annotations are lacking. Moreover, in non-fine-grained detection datasets, it is rare to encounter class confusion with more than two classes. Hence, the class confusion problem alleviated in this work is for two-class confusion. The two obstacles are visually shown in Fig. 1. Finally, BCNet did not update the annotations as training proceeds, which results in noisy weak pseudoannotations detracting from training all the time.

Building upon the above observations, we introduce the instance transformation network (ITNet), which includes three innovative modules: elastic cluster selection (ECS), label reassignment (LRA), and semisupervised elastic match (SSEM). Through ECS, weak pseudoannotations are selected by elastic threshold instead of hard threshold. Elastic threshold facilitates the selection of box annotations with more overlap of the instance. And LRA allows the ITNet to reassign the class-confused weak pseudoannotations by evaluating confusion coefficient and trained posterior knowledge. SSEM applies iterative updates, data augmentation, and pseudolabels' matching to continuously improve label quality.

To our knowledge, we are the first to present the original idea of ECS, LRA, and SSEM to LS-WSOD and the first to adopt LS-WSOD to RSIs. We have also conducted sufficient experiments to validate our model on the NWPU VHR-10.v2 [8] and DIOR [9] datasets. The obtained performance surpasses our baseline BCNet significantly.

In summary, the contributions of this article are as follows.

1) We propose ECS to obtain better weak pseudoboxes which alleviate part domination.
2) We propose LRA to correct class-confused labels, which alleviate two-class confusion in WSOD tasks.
3) SSEM is proposed to LS-WSOD, which continuously updates the weak pseudoannotations and manages to match pseudolabels' quality and quantity to reach higher performance.

## II. Related Work

### A. Weakly Supervised Object Detection

Most of the current works on WSOD are based on MIL [10], where they obtain region proposals in an image by unsupervised methods [11], [12] and treat all region proposals as instances and their image as a bag. Bilen and Vedaldi [1] proposed the weakly supervised deep detection network (WSDDN), which is the first revolutionary deep learning approach on MIL WSOD. Via the integration of the classification and detection scores of instances, WSOD loss is computed in comparison with the image-level annotations. WSDDN enabled WSOD in convolutional network, which achieved great performance improvement [1]. However, due to the lack of bounding box information, the network tends to detect only part of the object that is most discriminative, so much of the subsequent work revolves around solving the part domination in natural image set tasks, and meanwhile, other problems like instance ambiguity, memory consumption, augmentation consistency, etc., are studied intensively [2], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. And in the field of RSIs, numerous articles have made enhancements to the existing natural image methods considering RSIs' small objects, dense arrangements, and complex backgrounds that make tasks more difficult [3], [4], [5], [28], [29], [30], [31], [32], [33], [34]. Of late, Feng et al. [5] designed a global context-aware enhancement (GCAE) module, a dual-local context residual (DLCR) module, and an effective adaptive-weighted refinement loss to tackle the inaccurate detection and dense arrangement of objects. Yao et al. [31] proposed a dual-contextual instance refinement (DCIR) strategy and progressive contextual instance refinement (PCIR) in order to highlight the whole objects, filter out confusing negative samples, and meanwhile maintain the diversity of training samples. Wang et al. [33] proposed a multiple instance graph learning framework with a spatial graph vote mechanism and a parameter-free appearance graph instance mine strategy, which made it to pursue all possible object instances appearing in an image. Feng et al. [34] focused on adversarial dropout-activation to encourage the detection model to activate the entire object rather than focus on the most discriminative parts and the consistency and robustness when spatial transformation is employed. Shamsolmoali et al. [32]

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applied the concept of weak supervision to fully supervised object detection in RSIs. They proposed a novel architecture called the multipatch feature pyramid network (MPFP-Net) to address the issues of uneven scale, complex background, and missing object parts. Qian et al. [55] proposed the pseudoinstance soft label mining (PISLM) module to solve part domination and misclassification induced by copying neighbor instances' pseudolabels. Tan et al. [54] proposed the weakly supervised-oriented detector (WSODet) to achieve predict-oriented bounding box with only image-level labels.

Our work is most closely related to OICR [2] as our weak pseudoannotations are obtained from the output of OICR.

### B. Semisupervised Object Detection

The objective of semisupervised object detection (SSOD) is to utilize a large amount of unlabeled data and a small amount of labeled data to train the network for better performance. Depending on the method, SSOD can be divided into two types: consistency-driven [7], [35], [36], [37] and pseudoannotation-labeling-driven [38], [39], [40], [41], [42]. For RSIs, Chen et al. [43] extended the generative adversarial network (GAN) framework into object detection to generate fake samples. Zhang et al. [44] proposed SSOD-RS which works by augmentation-driven consistency regularization with OF-mixup and self-training. Wang et al. [45] introduced a semisupervised learning framework for SAR object detection, using consistent augmentation and label propagation.

Our work is also based on the study of augmentation consistency, whereby the data are augmented using the consistency principle in the label update of the SSEM module.

### C. Low-Shot WSOD

LS-WSOD is a recently proposed field where the network will conduct object detection with a few strong annotations and a large number of weak annotations. Yan et al. [46] presented EM algorithms for object detection, applicable to both weakly supervised and semisupervised settings, which is trainable when strong annotations and weak annotations both exist. Wang et al. [47] developed their strategy on Faster R-CNN by training the weakly supervised and semisupervised data with curriculum learning. The BCNet attempted to learn a transfer function to correct those incomplete boxes into complete ones with a two-branch two-path network [6]. Biffi et al. [48] introduced a novel online annotation module (OAM) that learns to generate annotations on the fly and thus affords concurrent training for fully supervised object detection.

Our work is most related to the BCNet which we introduce as a baseline. However, implemented for LS-WSOD, BCNet has three obvious shortcomings. First of all, the weak pseudoannotations are roughly obtained from OICR by hard threshold filtering, which does not consider the output distribution of the weakly supervised network and thus loses plenty of high-quality output. Second, BCNet is not designed to solve class confusion. Though low-shot strong annotations marked the category of instances, wrong confused weak pseudoannotations are always wrong instead of being rectified during training. Finally, BCNet does not take full advantage of the output on weak pseudolabeled data to refine the training data's label as semisupervised learning leverages the nonlabeled data.

## III. Proposed Method

As shown in Fig. 2, the ITNet's two-path network consists of a convolutional backbone as well as two parallel branches: correction branch and refinement branch.

Besides the two-path network aforementioned, the ECS is applied to generate improved weak pseudoannotations, the LRA module is plugged into the correction branch to correct class-confused labels, and the SSEM is employed to update the weak pseudoannotations to semi-pseudoannotations.

We employ BCNet's two-path network [6] as a baseline. Thus, we introduce two-path network first.

*Two-Path Network:* The purpose of the two-path network is to learn the ability to correct part-dominated boxes into complete boxes. The ITNet is trained on the low-shot weakly supervised training set $\mathcal{I}$, which is divided into two subsets, $\mathcal{I}_\alpha$ and $\mathcal{I}_\beta$. Samples in $\mathcal{I}_\alpha$ have both weak pseudoannotations and strong annotations, and samples in $\mathcal{I}_\beta$ have only weak pseudoannotations. Weak pseudoannotations are selected from the output of OICR [2] which is trained on only image-level labels. And strong annotations are obtained by manually labeling. Moreover, weak–strong box pairs are generated in $\mathcal{I}_\alpha$ through IoU greedy matching, which establishes a one-to-one mapping from weak pseudobox to strong box. The weak pseudoboxes and strong boxes are denoted as $b_w$ and $b_s$ relatively which indicate the boxes' information by $(x, y, w, h)$, and their one-hot class labels are denoted as $y_w$ and $y_s$. Semi-pseudoannotations and their augmented data generated by SSEM will join in $\mathcal{I}_\alpha$ after the first training stage, which will be discussed in Section III-C.

In order to make use of the weak pseudo- and strong annotations, two-path network is constructed as two branches and two paths: correction branch and refinement branch, and weak–strong mixed path $\alpha$ and only weak path $\beta$. The first-stage overall loss $L$ of ITNet is

$$\begin{aligned} L &= L_\alpha + L_\beta \\ &= L_{\alpha_C} + L_{\alpha_R} + L_{\beta_C} + L_{\beta_R} \end{aligned} \tag{1}$$

where

$$\begin{aligned} L_{\alpha_C} &= L_{\text{cls}}\left(p^\alpha, y_w^\alpha\right) + L_{\text{reg}}\left(t^\alpha, b_s^\alpha\right) \\ L_{\alpha_R} &= L_{\text{cls}}\left(p'^\alpha, y_w'^\alpha\right) + L_{\text{cls}}\left(p'^\alpha, y_s^\alpha\right) + L_{\text{reg}}\left(t'^\alpha, b_s^\alpha\right) \\ L_{\beta_C} &= L_{\text{cls}}\left(p^\beta, y_w^\beta\right) \\ L_{\beta_R} &= L_{\text{cls}}\left(p'^\beta, y_w'^\beta\right) + L_{\text{reg}}\left(t'^\beta, b_w'^\beta\right). \end{aligned} \tag{2}$$

We adopt cross entropy loss as $L_{\text{cls}}$ and the smooth$_{L1}$ loss as $L_{\text{reg}}$ as FRCNN [49] does. Accordingly, $p$ is the correction branch's output prediction score, $p'$ is the refinement branch's output prediction score, $t$ is the learned correction offset from weak boxes to strong boxes, and $b'_w$ and $y'_w$ are the transformed weak pseudobox and class from correction branch. The mixed path $\alpha$ and weak path $\beta$ let their respective data flow through the two-path network at each batch, thus generating their respective losses. The ability to correct from incomplete to

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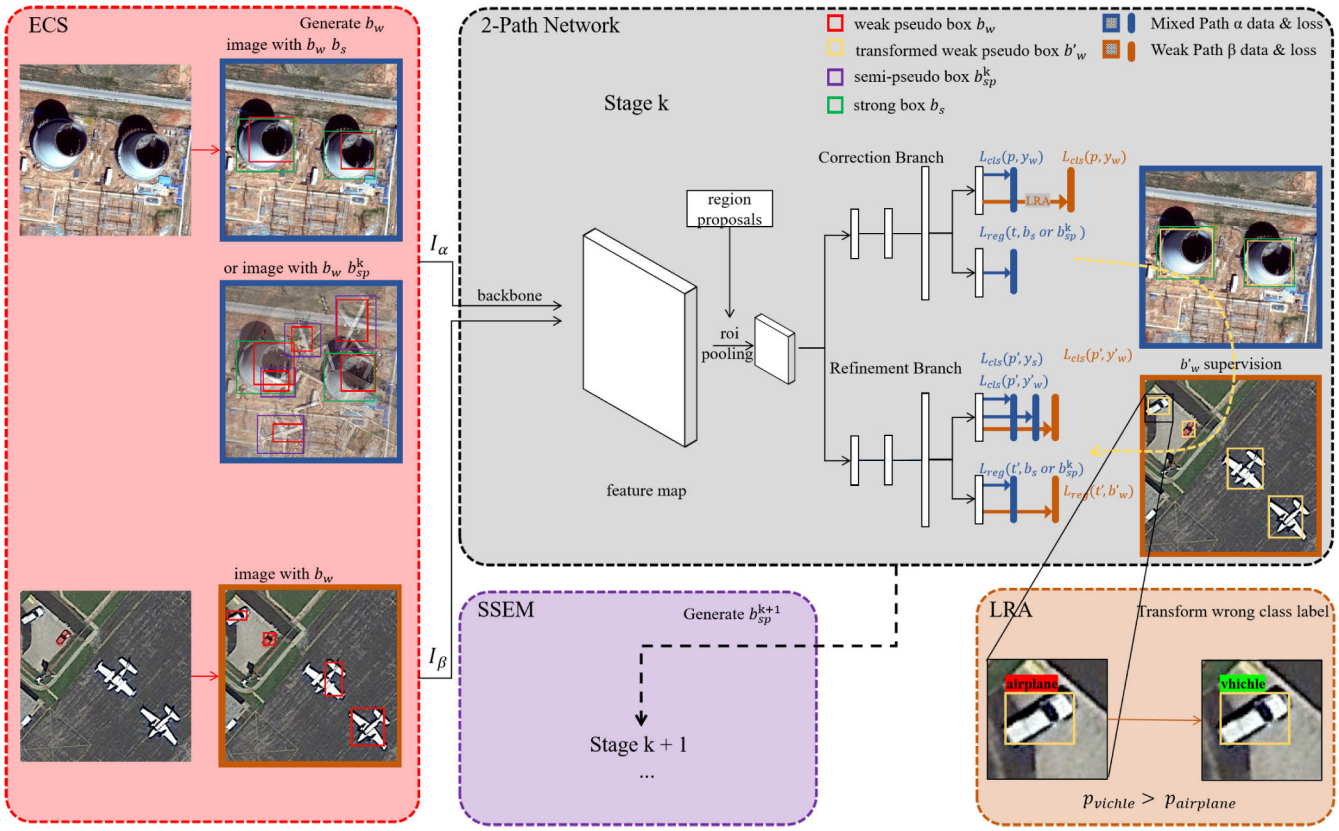



Fig. 2. Overview of our ITNet, which consists of four major parts: two-path network, ECS, LRA, and SSEM. The red, yellow, purple, and green bounding boxes in samples denote weak pseudoboxes $b_w$, transformed weak pseudoboxes $b'_w$, semi-pseudoboxes $b^k_{sp}$, and strong boxes $b_s$. The blue and orange colors indicate the different data and losses of the two paths. Each of the path of each batch has one image passing through the network in parallel. The two-path network is designed to train transforming ability from weak boxes to strong boxes by a two-path and two-branch structure. The ECS improves the quality of $b_w$ through an elastic and cluster-concerned selection strategy. The LRA could help network to alleviate two-class confusion by reassigning the confused class labels. The SSEM works by stages to continuously update labels, augment data, and match the quality and quantity of pseudolabels, which makes full advantages of strong labeled data and weak labeled data.

complete boxes is thus learned by weak–strong bounding box pairs. $\text{smooth}_{L1}$ loss is defined as follows:

$$\text{smooth}_{L1}(x) = \begin{cases} 0.5\ x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{3}$$

where $\text{smooth}_{L1}$ loss is a popular loss function used in regression tasks. It is a combination of $L1$ (absolute error) and $L2$ (squared error) loss functions, which helps to overcome some of their individual limitations.

The two-path network manages to utilize mixed data $\mathcal{I}_\alpha$ and only weak data $\mathcal{I}_\beta$. In correction branch, information of strong–weak pair in $\mathcal{I}_\alpha$ is learned to form the ability to transfer wrong instances' annotations. Furthermore, in the refinement branch, data with only weak pseudoboxes are also utilized under the supervision of $b'_w$. With better coverage of the object, $b'_w$ helps to leverage only weak labeled data of $\mathcal{I}_\beta$.

### A. Elastic Cluster Selection

Through ECS, weak pseudoannotations are selected from the output of OICR. According to Algorithm 1, for every output prediction result $I_i$ of OICR that has been NMSed in $I_O = (I_1, I_2, \ldots, I_i, \ldots, I_N)$, the boxes in $I_i$ having the highest prediction score $\widehat{P}_{ci}$ of class $c$ are picked as $\widehat{b}_c$. Elastic score $P^{\text{ela}}_{ci}$ is introduced to select high-quality output boxes from the rest of $I_i$

$$P^{\text{ela}}_{ci} = R_e \widehat{P}_{ci} \tag{4}$$

where the elastic rate $R_e$ is a hyperparameter. $\widehat{b}_c$ and boxes that meet the score requirements will be counted as weak pseudoannotations.

The cluster IoU threshold $I_e$ is applied as a hyperparameter to further mine the data from the bounding boxes of the clustered output. This continues filtering the bounding boxes in the non-NMS output $I'_O = (I'_1, I'_2, \ldots, I'_i, \ldots, I'_N)$, which allows more complete bounding boxes to be uncovered.

### B. Label Reassignment

Aiming to correct the wrong weak pseudoannotations caused by two-class confusion in WSOD, LRA is employed to transform the instances' category. The LRA module is plugged into the correction branch. In path $\alpha$, the class-confused labels in $b_w$ can be corrected directly because of the reference of strong annotations in correction branch. In path $\beta$, the class-confused labels in $b_w$ will be reassigned with a certain probability $p$, which is related to a confusion coefficient $m_{AB}$

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**Algorithm 1** Elastic Cluster Selection

**Input:** OICR inference result with NMS $I_O = (I_1, I_2, \ldots, I_i, \ldots, I_N)$ and without NMS $I'_O = (I'_1, I'_2, \ldots, I'_i, \ldots, I'_N)$, image label vectors $Y^{img} = (y_1^{img}, y_2^{img}, \ldots, y_i^{img}, \ldots, y_N^{img})$, elastic rate $R_e$, cluster IoU $I_e$

**Output:** weak pseudoannotations $b_w$ for instance transformation net

```
for I_i in I_O; do
    for class c in y_i^img do
        Pick b̂_c that has the top score P̂_ci in I_i, add P̂_c to B_WEAK;
        for b_ck ∈ rest of I_i do
            if P_cik > R_e P̂_ci then
                add b_ck to B_WEAK
            end
        end
        for b'_ck ∈ rest of I'_i do
            if IoU(b'_ck, B_WEAK) > I_e and
            P'_cik > R_e P̂_ci then
                add b'_ck to B'_WEAK
            end
        end
    end
end
Return b_w = B_WEAK ∪ B'_WEAK
```

and the training progress Iter

$$p(m_{AB}, \text{Iter}) = \frac{\text{Iter}}{\text{Total_iter}} m_{AB}^3 \tag{5}$$

where confusion coefficient $m_{AB}$ is introduced to describe the confusion level of category $A$ to category $B$. The higher the number of samples in which they are simultaneously present in a sample, the higher the confusion coefficient

$$m_{AB} = \frac{N_{AB}}{N_A} \tag{6}$$

where $N_A$ denotes the number of samples in which category $A$ appears and $N_{AB}$ represents the number of samples in which categories $A$ and $B$ both appear.

As an illustration, Algorithm 2 shows that in a triple classification problem, classes $A$, $B$, and $C$ exist, classes $A$ and $B$ are confused. For an instance $k$, its original one-hot weak pseudolabel is $y_k^\beta = [y_A, y_B, y_C]$. If $y_B = 1$, then class $A$ may be confused by class $B$, which means the confusion coefficient is $m_{AB}$. $y_k^\beta$ will be reassigned to $y_{\text{kLRA}}^\beta = [y_B, y_A, y_C]$ according to a certain probability $p(m_{AB}, \text{Iter})$. Note that the ability to reassign confused labels is learned from strong annotations in path $\alpha$, which is especially unstable in the early training procedure. So the exchange chance is low at the beginning of the training. What is more, the relationship between the confusion coefficient and training difficulty is nonlinear; thus, we introduce a cubic for the confusion coefficient $m_{AB}$.

After the reassignment, the validity of the exchange needs to be checked. Beforehand, the ability to recognize confused categories has been learned in path $\alpha$, so the validity of $y_{\text{kLRA}}^\beta$ depends on the category confidence $p^\beta = [p_A, p_B, p_C]$ of correction branch output in path $\beta$. If $p_A > p_B$, then the validity is proved and loss $L_{\beta_C}$ will be calculated as label $y_{\text{kLRA}}^\beta$; otherwise, the validity is disproved and loss $L_{\beta_C}$ will be calculated as label $y_k^\beta$ in this batch

$$L_{\beta_C} = \begin{cases} L_{\text{cls}}\left(p^\beta, y_{\text{kLRA}}^\beta\right), & \text{if } p_A > p_B \\ L_{\text{cls}}\left(p^\beta, y_k^\beta\right), & \text{otherwise.} \end{cases} \tag{7}$$

**Algorithm 2** Label Reassignment

**Input:** instance $k$'s weak pseudoclass labels $y_k^\beta = [y_A, y_B, y_C]$, category confidence $p^\beta = [p_A, p_B, p_C]$ from correction branch output, confusion coefficient $m_{AB}$, training progress $Iter$, total iterations $Total_iter$

**Output:** corrected weak pseudolabels, loss $L_{\beta_C}$

```
compute probability p(m_AB, Iter) = (Iter / Total_iter) m_AB^3
if instance k meets reassignment requirement then
    reassign one-hot weak pseudolabel y_k^β to y_kLRA^β = [y_B, y_A, y_C]
    compute loss L_βC as equation 7
    Return corrected label y_kLRA^β, loss L_βC
else
    compute loss L_βC = L_cls(p^β, y_k^β)
    Return original label y_k^β, loss L_βC
end
```

### C. Semisupervised Elastic Match

SSEM includes semisupervised mechanism and elastic match. Semisupervised mechanism is applied to our ITNet based on semisupervised learning for object detection [7], [50]. Similar to unlabeled data, weak pseudolabels can also be updated under the guidance of strong labels, which is based on self-training (via pseudolabel) and the augmentation-driven consistency regularization.

Multistage training is adopted to semisupervised mechanism. The first stage is the long-epoch stage, and the remaining stages are short-epoch stages. In the first stage, the ITNet is trained by weak pseudoannotations and strong annotations. Then, weak pseudoannotations are mined and updated to semi-pseudoannotations by hard threshold method. At the rest of the stages, our ITNet is trained by semi-pseudoannotations, remaining weak pseudoannotations and strong annotations.

In detail, referring to Fig. 3, after the first stage of a long-epoch training, the predicted output of ITNet on $\mathcal{I}_\beta$ is subjected to an elastic mean value $\mu$, and the compliant output will be updated to semi-pseudolabels. Meanwhile, strong augmentation strategies (e.g., mixup and moisac) are used on the semi-pseudolabeled data and strongly labeled data. Semi-pseudolabeled and strongly labeled data will be augmented together following a one-to-one, pseudo-to-strong strategy and be noted as $b_{sp}^k$ and $y_{sp}^k$ ($k = 2$ in this example) to train the $k$th short-epoch stage. In practice, the two-label augmented data

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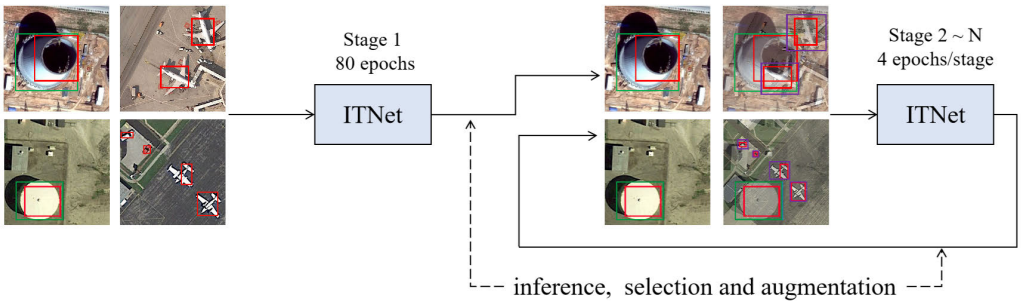



Fig. 3. Illustration of semisupervised mechanism. The first long-epoch stage ($k = 1$) is trained by weak pseudolabels $b_w$ and $y_w$ and strong labels $b_s$ and $y_s$. Then, compliant data in $\mathcal{I}_\beta$ will be mined, and its labels will be updated by hard threshold filtering. Moreover, updated data will be augmented together with strong data to form semi-pseudolabels $b_{sp}^{k+1}$. Then, in the next short-epoch stage, the ITNet is trained by $b_{sp}^{k+1}$, $b_s$, and the rest of $b_w$ and their class labels.

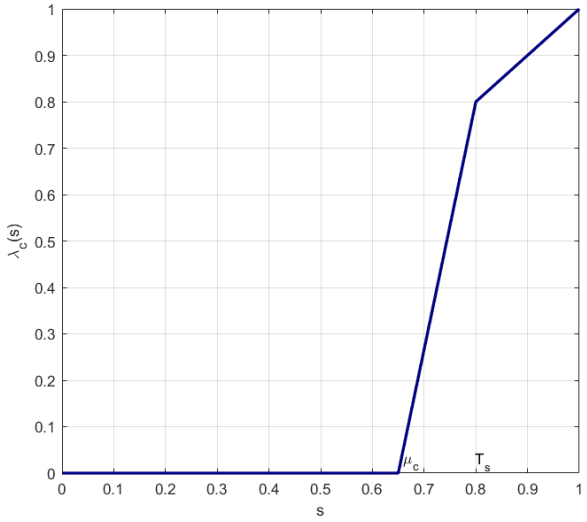



Fig. 4. Diagram of $\lambda_c(s)$, which shows the relationship between the semi-pseudolabel score $s$ and its loss coefficient $\lambda_c(s)$.

$b_{sp}^k$ is trained with the corresponding loss $L_{\alpha_C}^k$ being multiplied by a parameter $\lambda$ shown as follows:

$$L_{\alpha_C}^k = \lambda\left(L_{\text{cls}}\left(p^\alpha, y_{sp}^{\alpha k}\right) + L_{\text{reg}}\left(t^\alpha, b_{sp}^{\alpha k}\right)\right). \tag{8}$$

Under normal circumstances, $\lambda$ remains fixed. However, this can result in a mismatch between the quality and quantity of pseudolabels. Conventional hard-threshold filtering methods only account for the quality of pseudolabels, while quantity remains inadequate. To overcome this issue, we propose the elastic match that considers the characteristics of LS-WSOD and introduces a flexible approach in addition to the hard threshold. The elastic match enhances the quantity of pseudolabels while maintaining the quality and stability of training via a variable loss coefficient $\lambda(s)$. We designate the class with the highest score in $p^\alpha$ as $c^*$; then, $s$ is $p_{c^*}^\alpha$. We compute an elastic mean $\mu$, which enables the participation of pseudolabels with scores ranging from $\mu$ to $T_s$ (a hyperparameter for SSEM) in training while restricting their generated loss by $\lambda(s)$. The elastic mean $\mu$ is obtained by weighing the scores generated by pure WSOD (such as OICR) and ITNet, which assess the difficulty of the LS-WSOD task. The elastic mean

**Algorithm 3** Semisupervised Elastic Matching

**Input:** number of stage N,
training stage k's model $\mathcal{M}^k$
strong annotations $b_s$ and $y_s$
only weak annotations $b_w$ and $y_w$
semi-pseudoannotations $b_{sp}^k$ and $y_{sp}^k$
**Output:** trained model $\mathcal{M}^N$

```
for training stage from k to N do
    if k = 1 then
        set b_s and y_s as I_α, set b_w and y_w as I_β^k
        train model M^k on I_α and I_β^k
        generate μ^{k+1}, b_sp^{k+1} and y_sp^{k+1}, I_β^{k+1} using
          Algorithm 4
    else
        set b_s and y_s, b_sp^k and y_sp^k as I_α,
        train model M^k on I_α and I_β^k with elastic
          match
        generate μ^{k+1}, b_sp^{k+1} and y_sp^{k+1}, I_β^{k+1} using
          Algorithm 4
    end
    k++
end
Return M^N
```

value $\mu_c$ for class $c$ is defined as follows:

$$\mu_c = \frac{(1 - p_s)\mu_{cw} + (1 + p_s)\mu_{cs}}{2}, \quad (\text{shots} > 0) \tag{9}$$

where $p_s$ is the proportion of strong label data in the training set, $\mu_{cw}$ is the average value of ECS threshold in pure WSOD task, and $\mu_{cs}$ is the average value of ECS threshold in ITNet. It is worth noting that the output of ITNet has not been filtered by $\mu_s$, but only used in the calculation of $\mu$.

Outputs with the $s$ less than $\mu$ will not be selected as semi-pseudolabels. Pseudolabels with scores between $\mu$ and $T_s$ will participate in training with a smaller loss weight $\lambda$. Finally, for continuity of $\lambda$, when $s$ is greater than $T_s$, the $\lambda$ is equal to the pseudolabel score. Thus, we can obtain the $\lambda_c(s)$ function as shown in Fig. 4 and set $\lambda(s)$ as $\lambda_{c^*}(s)$.

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**Algorithm 4** Semisupervised Label Generation

**Input:** training stage k's model $\mathcal{M}^k$,
strong annotations $b_s$ and $y_s$,
only weak data $\mathcal{I}_\beta^k$,
semi-pseudoannotations $b_{sp}^k$ and $y_{sp}^k$
**Output:** updated $b_{sp}^{k+1}$ and $y_{sp}^{k+1}$, updated $\mathcal{I}_\beta^{k+1}$, $\mu^{k+1}$

$\mathcal{M}^k$ inference on $\mathcal{I}_\beta^k$
compute elastic mean $\mu^{k+1}$ by equation 9
apply hard threshold filtering on inference result of $\mathcal{I}_\beta^k$, with the threshold of $\mu^{k+1}$
strong augmentation for strong labeled data and filtered data and set the augmented data as $b_{sp}^{k+1}$ and $y_{sp}^{k+1}$
set the remaining data of $\mathcal{I}_\beta^k$ as $\mathcal{I}_\beta^{k+1}$
**Return** $b_{sp}^{k+1}$ and $y_{sp}^{k+1}$, $\mathcal{I}_\beta^{k+1}$, $\mu^{k+1}$

Then, $\lambda_c(s)$ can be defined as (10), and (8) can be redefined as (11)

$$\lambda_c(s) = \begin{cases} 0, & 0 \le s \le \mu_c \\ \dfrac{T_s}{T_s - \mu_c}(s - \mu_c), & \mu_c < s < T_s \\ s, & T_s \le s \le 1 \end{cases} \tag{10}$$

$$\begin{cases} c^* = \arg\max_c \ p_c^\alpha \\ L_{\alpha_c}^k = \lambda_{c^*}(s)\left(L_{\text{cls}}(p^\alpha, y_{sp}^{\alpha k}) + L_{\text{reg}}\left(t^\alpha, b_{sp}^{\alpha k}\right)\right). \end{cases} \tag{11}$$

In summary, the algorithmic representation of the SSEM method is as Algorithm 3.

## IV. Experiments

### A. Datasets and Evaluation Metrics

To evaluate the proposed ITNet, we use NWPU VHR-10.v2 with ten categories and DIOR with 20 categories for object detection. The NWPU VHR-10.v2 has 879 images for training and 293 images for testing. The DIOR has 11 725 images for training and 11 738 images for testing. We select all training set images to train our model with different shots. For example, we choose 200 strong annotations and 11 725 weak pseudoannotations of DIOR to train ITNet on the ten-shot weakly supervised task. The experiments are carried out on five different random folds of ten-shot dataset. Mean precision (mAP) is employed to evaluate our method, which is used to test the detection performance of our model on the test dataset. mAP is computed based on an IoU threshold of 0.5 between predicted bounding boxes and ground truth.

### B. Implementation Details

*1) Convolutional Backbone:* VGG16 [51] is our primary experiment backbone, because most of the WSOD tasks are based on it. In addition, we also give results on ResNet101 [52] for reference.

*2) Experiment Environment:* Experiments are conducted using an SGD optimizer on an RTX 2080Ti GPU. For the hyperparameter configuration, the elastic rate $R_e$, the cluster IoU $I_e$, and the hard threshold $T_s$ for SSEM are all set as 0.8. For the training configuration, the initial learning rate is 0.001, the batch size is 1, the training momentum is 0.9, and the weight decay is 0.0005. Training images will be resized to scales of (400, 600, 750) and will be randomly horizontally flipped for augmentation, especially semi-pseudolabeled data will be augmented by the moisac and mixup method. The first stage of ITNet will be trained for 80 epochs, and the remaining stages are four epochs each, for a total of four stages. The nonmaxima suppression (NMS) is adopted with an IoU threshold of 0.3.

*3) Hyperparameter Study:* Fig. 5 reports the mAP performance of different values of three different hyperparameters. By the grid search method, it is found that the optimal setting of hyperparameters is $R_e = 0.8$, $I_e = 0.8$, and $T_s = 0.8$.

TABLE I
Effect of Each Module to mAP (%) on DIOR

| ITNet on VGG16 | | | | | | |
|---|---|---|---|---|---|---|
| + Elastic cluster selection | | √ | | | √ | √ |
| + Label re-assignment | | | √ | | √ | √ |
| + Semi-supervised elastic match | | | | √ | | √ |
| mAP(%)(10-shot) | 36.2 | 38.2 | 37.1 | 39.3 | 39.7 | 42.3 |

### C. Ablation Experiments

For the three modules we propose, ECS, LRA, and SSEM, ablation experiments in the ten-shot case are performed. The experiments demonstrate the performance improvement of the individual modules as well as when combined as shown in Table I.

*1) Elastic Cluster Selection:* First, the ECS, by adding ECS to BCNet, we improved the mAP performance from 36.2% to 38.2%. It shows that although the ECS method increases the number of boxes of $b_w$, it improves the quality of weak pseudoannotations. Among the additional annotations, most of them have better coverage of instance, and rest of them can be tolerated by the ITNet as moderate noise.

*2) Label Reassignment:* The performance improvement of BCNet when LRA is introduced alone is improved (from 36.2% to 37.1%). However, when ECS and LRA are introduced in parallel, the performance improvement is more significant in comparison (from 36.2% to 39.7%). This is because the ECS method inevitably introduces more class-confused, relatively low-scoring bounding boxes, while LRA just solves this problem. This confirms that instances which are confused by categories have a negative impact on training process, resulting in low test result scores. It further proves that the label confused by categories is only a category problem, while the quality of its bound box is valid. The output affected by category confusion is then more easily selected by the ECS and then corrected by the LRA. Therefore, the integration of ECS and LRA approach gives better performance.

*3) Semisupervised Elastic Match:* When SSEM is included only, ITNet's performance raised from 36.2% to 39.3%, indicating that BCNet's inability to update labels is a fairly obvious shortcoming. When ECS, LRA, and SSEM are added at the same time, the performance reached a maximum of 42.3%, indicating that the contributions of the three modules

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TABLE II

OUR ITNET AND OTHER MODELS INFERENCE COMPUTATIONAL PERFORMANCE ON THE DIOR DATASET [9]. PARAMS (IN MILLIONS) AND FLOPS (IN BILLIONS) REPRESENT THE NUMBER OF THE PARALLELS AND MULTIPLY ADDS. LAT SIGNIFIES INFERENCE LATENCY WITH BATCH SIZE 1

| Methods | mAP(%) | Params | Ratio | FLOPs | Ratio | GPU LAT(s) | Speedup | CPU LAT(s) | Speedup |
|---|---|---|---|---|---|---|---|---|---|
| WSDDN [1] | 13.3 | 268.6M | 1.99× | 2,696.1B | 6.70× | 2.21 | 6.31× | 2.23 | 18.58× |
| OICR [2] | 16.5 | 137.4M | 1.02× | **402.2B** | **1**× | **0.35** | **1**× | 0.86 | 7.17× |
| BCNet [6] | 36.1 | **134.7M** | **1**× | 869.2B | 2.16× | 0.53 | 1.51× | **0.12** | **1**× |
| Ours | **42.3** | **134.7M** | **1**× | 869.2B | 2.16× | 0.53 | 1.51× | **0.12** | **1**× |

TABLE III

FEW-SHOT PERFORMANCE (MAP%) BY VARYING NUMBER OF STRONG ANNOTATION SHOTS

| methods | 4-shot | 10-shot | 20-shot | Fully supervised |
|---|---|---|---|---|
| BCNet [6] | 28.0 | 36.1 | 40.9 | 54.5 |
| Ours | 31.9 | 42.3 | 44.5 | 54.5 |

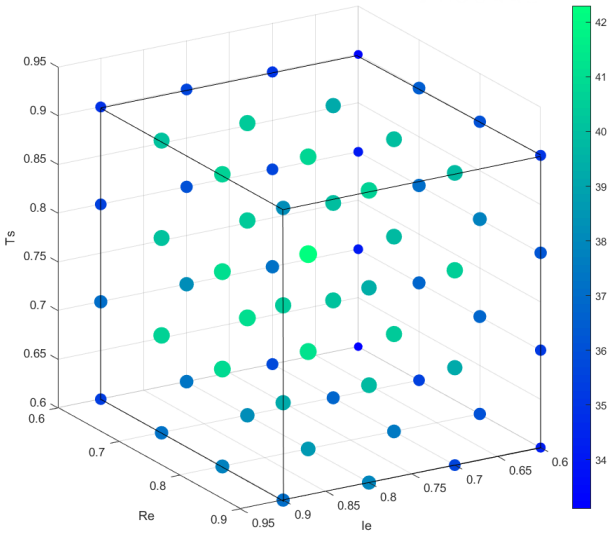



Fig. 5. Ablation study of different values of three hyperparameters on the DIOR dataset. The experiment is conducted using the grid search approach. The point with larger size and greener color indicates better performance.

are relatively independent, and there is no distinct overlap between the three functions.

### D. Visualization Analysis

Instance-level transformation ability includes the capability to correct both bounding boxes and categories. In this section, we will showcase ITNet's ability to perform instance-level corrections through two visualized data.

*1) Box Correction:* Fig. 6 provides a visualization of different pseudolabel acquisition methods. Compared to the hard threshold method, which serves as the baseline, the ECS is able to explore more pseudolabels but may also introduce noise. Additionally, the SSEM can suppress noise, meanwhile increasing the number of high-quality pseudolabels. Experimental results demonstrate that combining ECS and SSEM methods can effectively enhance both the quantity and quality of weak pseudolabels.

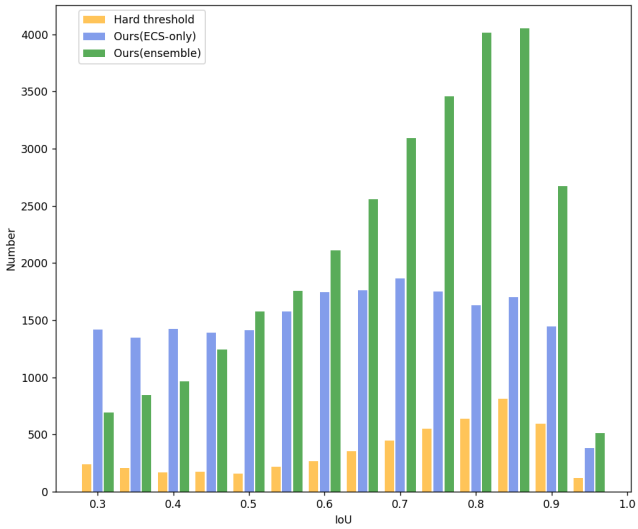



Fig. 6. Comparison of the IoU distribution of pseudolabels obtained from different methods using the DIOR dataset as the benchmark. The three kinds of pseudolabels are obtained from: 1) BCNet's hard threshold method; 2) ITNet's ECS-only method; and 3) ITNet's ensemble method from the last stage of SSEM.

*2) Class Correction:* From Fig. 7, the t-SNE [53] method is applied to feature vectors of BCNet and ITNet. It is evident that the feature vectors obtained by BCNet are still confused obviously, while those obtained by ITNet are relatively clustered within the class and dispersed between classes, making them more effective. Therefore, ITNet is better equipped to address the issue of category confusion, as deduced from the graph.

### E. Comparisons

Table II presents the computational performance of ITNet alongside other models on the DIOR dataset. The findings show that ITNet outperformed other models with the highest mAP performance (42.3%), without additional costs on model parameters and computational complexity than the baseline. Moreover, the CPU latency of ITNet is on par with the BCNet model, both achieving the best result of 0.12 s.

Table III shows the mAP performance comparison between our ITNet and BCNet of different shots on DIOR. Compared with BCNet, ITNet's mAP has improved by 3.9%, 6.2%, and 3.6% for four-shot, ten-shot, and 20-shot, respectively, and they both have the same performance of 54.5% in the fully supervised case. It is worth noting that the promotion of ITNet is the highest under ten-shot. First, by increasing the number

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TABLE IV

AVERAGE PRECISION (%) ON NWPU VHR-10.V2 TEST DATASETS OF DIFFERENT MODELS. THE PERFORMANCE DATA OF WSDDN, OICR, TCANET, SAENET, PISLM, AND WSODET ARE COLLECTED FROM [55] AND [54]. ONLY WEAKLY SUPERVISED LABELS MEAN THAT THE MODEL IS TRAINED BY USING ONLY IMAGE-LEVEL LABELS. LOW-SHOT WEAKLY SUPERVISED LABELS MEAN THAT BOTH IMAGE-LEVEL LABELS AND A SMALL AMOUNT OF STRONG LABELS ARE USED. TEN-SHOT CASE MEANS THAT EVERY CLASS ONLY HAS TEN IMAGES WITH STRONG LABELS (100 IMAGES IN TOTAL)

| Methods, Backbone | Airplane | Ship | Storage tank | Baseball diamond | Tennis court | Basketball court | Ground track field | Harbor | Bridge | Vehicle | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Only weakly supervised labels: | | | | | | | | | | | |
| WSDDN [1], VGG16 | 30.1 | 41.7 | 35.0 | 88.9 | 12.9 | 23.9 | 99.4 | 13.9 | 1.9 | 3.6 | 35.1 |
| OICR [2], VGG16 | 13.7 | 67.4 | 57.2 | 55.2 | 13.6 | 39.7 | 92.8 | 0.2 | 1.8 | 3.7 | 34.5 |
| TCANet [5], VGG16 | 89.4 | 78.2 | 78.4 | 90.8 | 35.3 | 50.4 | 90.9 | 42.4 | 4.1 | 28.3 | 58.8 |
| SAENet [34], VGG16 | 82.9 | 74.5 | 50.2 | 96.7 | 55.7 | 72.9 | **100.0** | 36.5 | 6.3 | 31.9 | 60.7 |
| WSODet [54], VGG16 | **95.3** | 75.6 | 81.9 | **98.0** | 20.9 | 56.7 | **100.0** | 29.8 | 5.1 | 48.1 | 61.3 |
| PISLM [55], VGG16 | 87.6 | 80.1 | 57.3 | 94.0 | 36.4 | **80.4** | **100.0** | 56.9 | 9.8 | 35.6 | 63.8 |
| Low-shot weakly supervised labels (10-shot): | | | | | | | | | | | |
| BCNet [6], ResNet101 | 61.5 | 85.1 | 89.9 | 88.1 | 71.3 | 71.2 | 98.0 | 61.3 | 9.7 | 58.8 | 69.5 |
| BCNet [6], VGG16 | 60.1 | **89.3** | 89.5 | 86.5 | 77.7 | 70.8 | 99.2 | 57.8 | 14.5 | 59.2 | 70.5 |
| Ours, ResNet101 | 78.2 | 86.2 | 83.9 | 85.5 | 77.6 | 65.8 | 98.4 | 69.9 | **24.0** | **59.6** | 73.6 |
| Ours, VGG16 | 89.0 | 86.1 | **90.2** | 82.2 | **78.2** | 79.3 | 99.6 | **78.0** | 15.7 | 56.2 | **75.3** |

TABLE V

AVERAGE PRECISION (%) ON DIOR TEST DATASETS OF DIFFERENT MODELS. THE PERFORMANCE DATA OF WSDDN, OICR, TCANET, SAENET, PISLM, AND WSODET ARE COLLECTED FROM [55] AND [54]. ONLY WEAKLY SUPERVISED LABELS MEAN THAT THE MODEL IS TRAINED BY USING ONLY IMAGE-LEVEL LABELS. LOW-SHOT WEAKLY SUPERVISED LABELS MEAN THAT BOTH IMAGE-LEVEL LABELS AND A SMALL AMOUNT OF STRONG LABELS ARE USED. TEN-SHOT CASE MEANS THAT EVERY CLASS ONLY HAS TEN IMAGES WITH STRONG LABELS (200 IMAGES IN TOTAL)

| Methods, Backbone | Airplane | Airport | Baseball field | Basketball court | Bridge | Chimney | Dam | Expressway service area | Expressway toll station | Golf field | - |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Only weakly supervised labels: | | | | | | | | | | | |
| WSDDN [1], VGG16 | 9.1 | 39.7 | 37.8 | 20.2 | 0.3 | 12.2 | 0.6 | 0.7 | 11.9 | 4.9 | - |
| OICR [2], VGG16 | 8.7 | 28.3 | 44.1 | 18.2 | 1.3 | 20.2 | 0.1 | 0.7 | 29.9 | 13.8 | - |
| TCANet [5], VGG16 | 25.1 | 30.8 | 62.9 | 40.0 | 4.1 | 67.8 | 8.1 | 23.8 | 29.9 | 22.3 | - |
| SAENet [34], VGG16 | 20.6 | **62.4** | 62.7 | 23.5 | 7.6 | 64.6 | 0.2 | 34.5 | 30.6 | 55.4 | - |
| WSODet [54], VGG16 | 32.2 | 53.3 | 66.5 | **76.6** | 0.1 | 57.1 | 0.1 | 0.1 | 0.4 | 42.8 | - |
| PISLM [55], VGG16 | 29.1 | 49.8 | **70.9** | 41.4 | 7.2 | 45.5 | 0.2 | 35.4 | 36.8 | **60.8** | - |
| Low-shot weakly supervised labels (10-shot): | | | | | | | | | | | |
| BCNet [6], ResNet101 | 26.8 | 46.9 | 61.8 | 35.4 | 10.2 | 58.9 | 13.8 | 37.1 | 23.6 | 38.2 | - |
| BCNet [6], VGG16 | 36.2 | 43.7 | 64.2 | 59.8 | 11.4 | 54.8 | 13.1 | **38.4** | 19.7 | 42.3 | - |
| Ours, ResNet101 | 41.9 | 47.2 | 66.0 | 50.2 | 13.2 | **68.9** | **13.9** | 36.0 | **43.2** | 31.7 | - |
| Ours, VGG16 | **53.0** | 45.6 | 68.3 | 75.5 | **13.7** | 68.8 | 13.8 | 35.5 | 42.6 | 30.6 | - |

| Methods, Backbone | Ground track field | Harbor | overpass | Ship | Stadium | Storage tank | Tennis court | Train station | Vehicle | Wind mill | mAP(%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Only weakly supervised labels: | | | | | | | | | | | |
| WSDDN [1], VGG16 | 42.4 | 4.7 | 1.1 | 0.7 | 63.0 | 4.0 | 6.1 | 0.5 | 4.6 | 1.1 | 13.3 |
| OICR [2], VGG16 | 57.4 | 10.7 | 11.1 | 9.1 | 59.3 | 7.1 | 0.7 | 0.1 | 9.1 | 0.4 | 16.5 |
| TCANet [5], VGG16 | 53.9 | 24.8 | 11.1 | 9.1 | 46.4 | 13.7 | 31.0 | 1.5 | 9.1 | 1.0 | 25.8 |
| SAENet [34], VGG16 | 52.7 | 17.6 | 6.9 | 9.1 | 51.6 | 15.4 | 1.7 | **14.4** | 1.4 | 9.2 | 27.1 |
| WSODet [54], VGG16 | **66.6** | 0.1 | 1.9 | 2.0 | 52.6 | 22.4 | 68.8 | 0.2 | 1.2 | 0.3 | 27.3 |
| PISLM [55], VGG16 | 48.5 | 14.0 | 25.1 | 18.5 | 48.9 | 11.7 | 11.9 | 3.5 | 11.3 | 1.7 | 28.6 |
| Low-shot weakly supervised labels (10-shot): | | | | | | | | | | | |
| BCNet [6], ResNet101 | 54.2 | 28.3 | 22.3 | 20.9 | 57.1 | 24.6 | 66.3 | 2.6 | 17.6 | 33.4 | 34.0 |
| BCNet [6], VGG16 | 56.9 | 26.5 | 26.8 | 21.0 | 56.1 | 26.9 | 68.5 | 2.4 | 20.2 | **34.0** | 36.1 |
| Ours, ResNet101 | 66.2 | **33.7** | 27.0 | 25.5 | **59.7** | 34.3 | 73.0 | 6.5 | 26.3 | 33.6 | 40.5 |
| Ours, VGG16 | **66.6** | 32.6 | **30.5** | **26.7** | 59.3 | **36.5** | **74.6** | 6.6 | **27.2** | **34.0** | **42.3** |

of shots after ten-shots, the performance gap between ITNet and BCNet will gradually narrow because the contribution of strong labeled data will gradually dominate the training contribution, thus making less contribution to the module improvement of the ITNet. Secondly, when the number of shots is less than 10, there is too little strong labeled data

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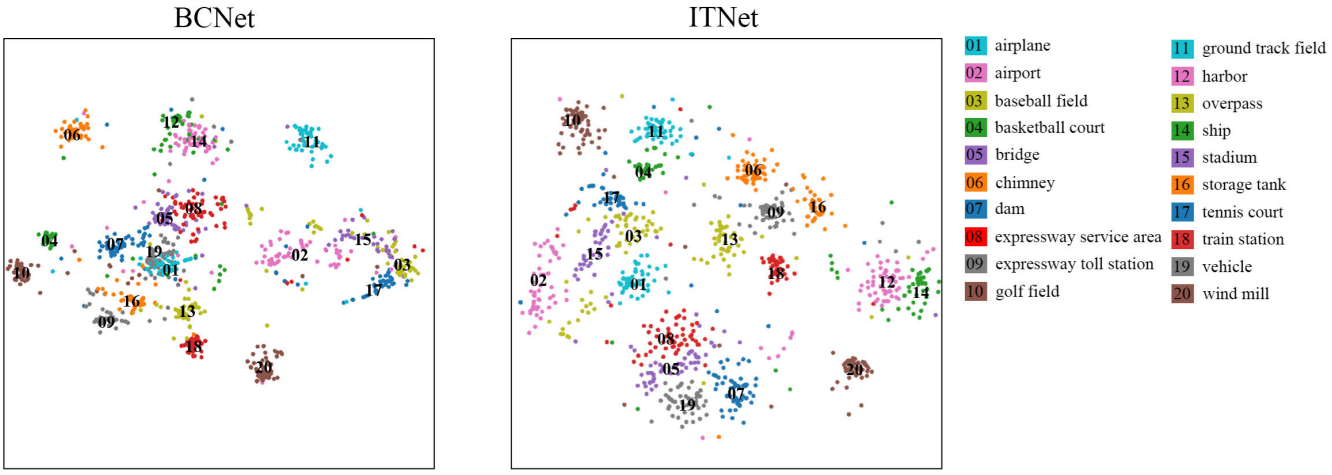



Fig. 7. t-SNE dimensional reduction visualization of objects' feature vectors on DIOR. Each category has 60 vectors participating in the visualization.

available, which makes the LRA and SSEM of ITNet not having enough strong labeled data as reference, resulting in insufficient performance improvement. Therefore, although the experimental data of different shots shows a certain degree of nonsmoothness, this can be reasonably explained and is in line with the module design principle of our ITNet.

In Table IV, we present the mAP performance of different models on the NWPU VHR-10.v2 test dataset. The table is divided into two sections, one for models trained using only weakly supervised labels and another for models trained using low-shot weakly supervised labels with ten-shot setting. Our proposed model with VGG16 backbone achieves the highest mAP of 75.3%, followed by our model with ResNet101 backbone (73.6%), BCNet with VGG16 backbone (70.5%), and BCNet with ResNet101 backbone (69.5%). Our VGG16 model outperforms other models in several categories, including airplane (89.0%), storage tank (90.2%), tennis court (78.2%), basketball court (79.3%), and harbor (78.0%).

Table V illustrates the mAP performance of our ITNet and other methods on DIOR. With ten-shot case (i.e., 200 strongly labeled images), our ITNet mAP performance already significantly exceeds that of the purely weakly supervised models: WSDDN (+29.0%), OICR (+25.8%), TCANet (+16.5%), SAENet (+15.2%), WSODet(+15.0%), and PISLM(+13.7%). In terms of BCNet as a benchmark, our model exceeds the mAP by 6.2%, for the same ten-shot case. Regarding single-class mAP performance compared to our baseline BCNet, our approach improves for most classes. Among the classes that have obviously improved performance are: airplane (+16.8%), baseball filed (+4.1%), basketball court (+15.7%), chimney (+14.0%), expressway toll station (+22.9%), ground track filed (+9.7%), harbor (+6.1%), storage tank (+9.6%), tennis court (+6.1%), and vehicle (+7.0%). Measured by the confusion coefficient, the 20 categories can be divided into two parts: nine with high confusion coefficients (greater than 0.3) and 11 with low confusion coefficients. The first part includes harbor, expressway service area, ship, overpass, expressway toll station, basketball court, stadium, tennis court, and airplane. The other part includes ground track field, bridge, baseball field, chimney, storage tank, vehicle, golf field, train station, dam, airport, and windmill. It can be figured that the mAP of nine former classes with high confusion coefficients improves by 8.3%, while 11 classes with low confusion coefficients improves by only 4.1%. This illustrates that our proposed ITNet is particularly effective for the confused classes, which is the result of contribution of the three modules ECS, LRA, and SSEM to the data mining and label updating.

It is also worth noting that our ITNet encounters performance degradation on golf field (−11.7%) and expressway service area (−2.9%). The two types of instances are composed of multiple parts, and the distribution of each part is relatively scattered. This results in the ECS not being able to mine better labels, and instead, the noise ECS introduces can cause performance degradation. And when backbone is replaced with ResNet101, airplane (−11.1%) and basketball court (−25.3%) also show a huge performance decrease, which leads to a decrease in mAP. This may be due to the fact that the ResNet is too deep to restore some small object information of these two classes during the network training. The other possibly problem is that the WSOD head is sensitive to model initialization and has instabilities that may back-propagate uncertain and erroneous gradients to the backbone, deteriorating visual representation learning [56].

In summary, the experimental comparison demonstrates the superiority of our approach over existing only weakly supervised and low-shot weakly supervised methods.

### F. Qualitative Results

Fig. 8 shows the successful (a) and (b) and unsuccessful (c) detection results of ITNet. Fig. 9 shows the comparison of several results of OICR, BCNet, and ITNet in the same image, respectively. Since the score distributions of the prediction results are different between the three networks, we normalize the scores of the three in order to present the results in a uniform manner. Notably, in Fig. 9, column 1 shows that ITNet is more complete in detecting targets. And column

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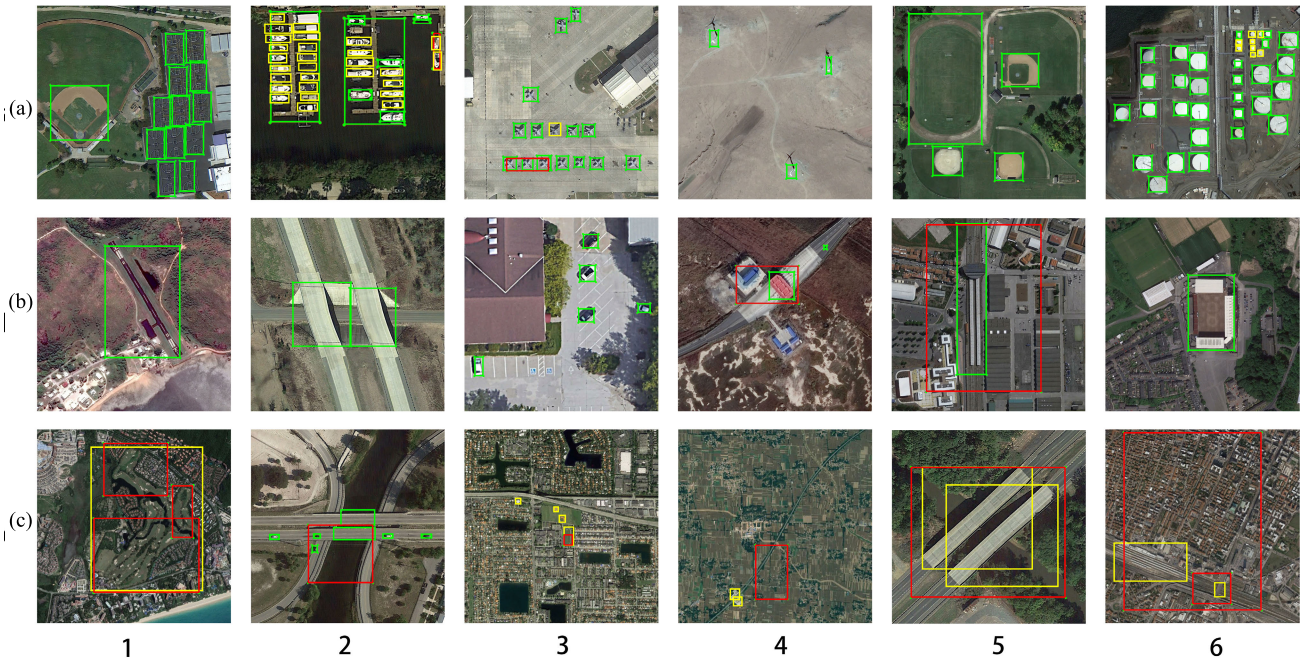



Fig. 8. Some detection results on the DIOR test set under 10-shot supervision (42.3% mAP). Row (a) and (b) show some relatively successful results and row (c) shows some relatively unsuccessful results. The results include successful cases with green rectangles, unsuccessful cases with red rectangles and missed cases with yellow rectangles.

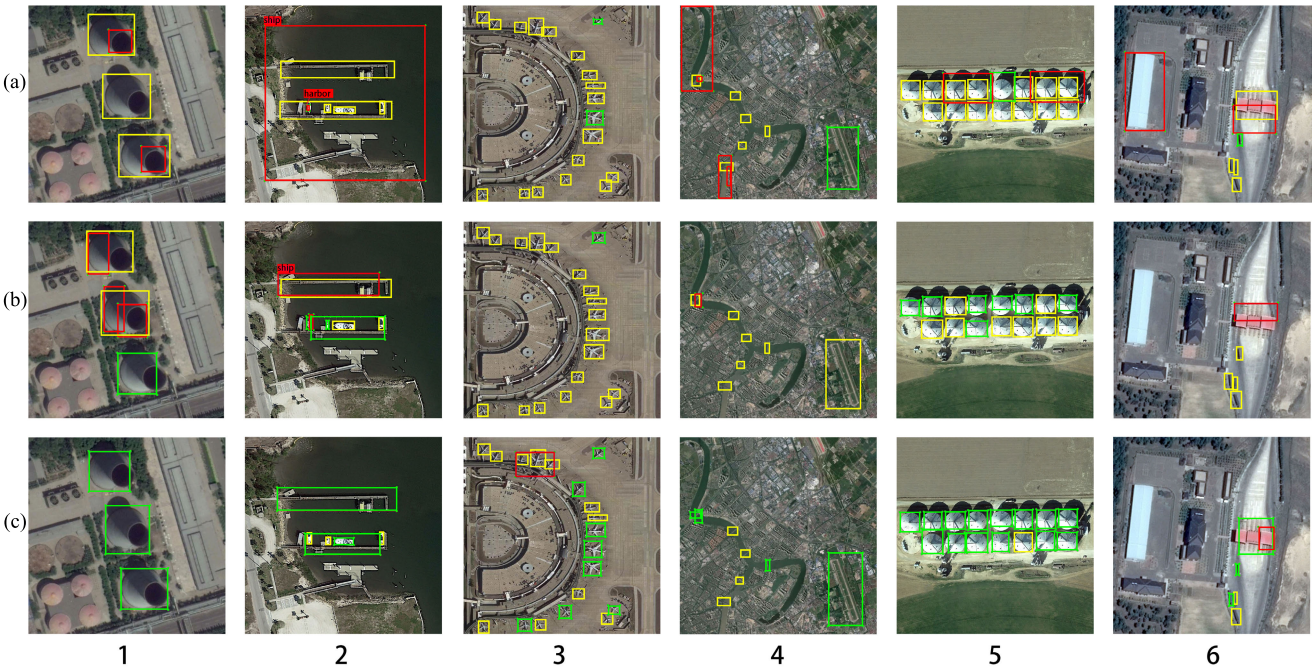



Fig. 9. Some detection results comparison on the DIOR test set between OICR, BCNet, and ITNet under the same ten-shot supervision. Results of (a) OICR, (b) BCNet, and (c) ITnet, relatively. The results include successful cases with green rectangles, unsuccessful cases with red rectangles, and missed cases with yellow rectangles. If the unsuccessful cases failed with recognizing the class, their red rectangles will be noted with the incorrectly recognized class in the upper left corner additionally.

2 shows that the ITNet is able to deal with the two-class confusion problem in WSOD. Columns 3–6 show that ITNet is successfully trained to detect more objects than OICR and BCNet in different scenes.

## V. Conclusion

The ITNet solves the problem when data are simultaneously labeled by a large number of weak labels and a small number of strong labels. Compared with weakly supervised methods and low-shot weakly supervised methods, the ITNet focuses more on the relationship between weak pseudolabels and strong labels so the two-class confusion and noisy weak pseudolabels problem could be alleviated by LRA and SSEM. With our approach, ITNet has not only transformed the part-dominated bounding boxes but also corrected the confused categories, which implements the transformation at the instance level. More importantly, the ITNet is more pertinent to practical application scenarios, where the annotation work

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is carried on progressively, or a large number of samples are annotated automatically, which has a broad application prospect.

However, ITNet is not an end-to-end network. Its weak pseudolabels are selected from the output of OICR. And in SSEM, weak pseudolabels are not updated in real time through training. Further work such as end-to-end LS-WSOD network and research of online label-updating will be worth investigating.

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